Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/05/2020
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Constrained Vapor Bubble Flight Experiment II – Binary Mixtures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2013	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2019 per NSSC information (Ed., 5/13/19) NOTE: End date changed to 3/31/2019 per NSSC information (Ed., 12/7/17)		
Key Personnel Changes/Previous PI:	July 2019 report: Jiaheng Yu is working on the project, Jack Keating is a graduate student working for the summer, and Alex Guo is an undergrad working for the summer. July 2018 report: Jiaheng Yu is working on the project and Sarah Dalakos is an undergrad working for the summer. July 2017 report: Graduate student Akshay Kundan graduated; Graduate student Thao Nguyen graduated; Graduate student Jiaheng Yu joined the group.		
COI Name (Institution):	Wayner, Peter Ph.D. (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX13AQ78G		
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Performance Goal Text:			

Task Description:	The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics, and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for microgravity conditions. The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates. The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, micro-electromechanical systems (MEMS) devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free. The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth.
	FINAL REPORTING JANUARY 2020 SUMMARY: During the course of this project we have run the CVB-2 experiments on the ISS and have analyzed the data from both CVB-2 and CVB-1 flight experiments. We have focused on as many aspects of those experiments as we could, at least making a pass through most of the data we had collected. We have used some of the time to rebuild our ground-based CVB system and refine our image analysis procedures to extract more data from the fringe images and reliably determine values for the in-situ Hamaker constant of pure fluids and fluid mixtures. We have given 32 presentations related to this project, graduated two PhD students, partially supported two others, supported six undergraduates doing research projects, and have published 13 papers come Spring 2020. We continue to analyze data, to present new information and models at as many conferences as we could reasonably attend, and have been pursuing some ideas on wicking structures and surface modification techniques using atomic layer deposition and atom-transfer-radical polymerization that would enhance the rates of heat transfer. MAJOR PROJECT HIGHLIGHTS: The major highlights from this project are listed below and the reference details are given at the end of the report. 1. We analyzed the high heat input experiments from CVB-1 and showed that the decrease in performance of a wickless heat pipe like CVB in microgravity is not related to dryout or boiling limitations but instead results from flooding of the hot end. This result was never before seen in heat pipes and arises from a competition between Marangoni forces at the hot end and capillary forces at the cold end. It appears as dryout in the temperature profile so only visualization allowed us to see the real cause. 2. Based on the results from #1 we presented in Physical Review Letters paper, the group won the 2015 ISS Compelling Results Award. [Ed. note: see FY2016 report Bibliography for details] 3. We compiled and analyzing data from the flight experiments associated with CVB-2. CVB-2, when it was working reliably, performed as expected. The hypothesis that adding a small amount of a component with a higher boiling point and higher surface tension could counteract the Marangoni flooding issues seen in CVB-1 was proven. This worked even though the pentane-isohexane fluid used, formed an ideal mixture. 4. Heat transfer coefficients in the evaporator section of CVB-2 were nearly double those of CVB-1. This had to do with breaking the Marangoni problem and we also were able to show that the composition of fluid in CVB-2 varied along the length of the device. Assuming an ideal mixture of pentane and isohexane, the fluid near the heater end was nearly pure isohexane while the nominal mixture was 94% pentane – 6% isohexane.

Task Progress:

5. We analyzed the effect of condenser temperature on the evaporator performance of CVB-2. Even though the condenser is located over 30 mm away from the heater, altering the condenser temperature had a great effect on the heat pipe's performance and the composition near the heater end.

6. Similar behavior was then observed in older CVB-1 data. The condenser temperature, even for the pure fluid, exerts a tremendous influence on the processes occurring at the heater end and can nearly double the deleterious effects of Marangoni flooding as the condenser temperature was lowered dramatically below ambient.

7. The high heat input data on CVB-1 has led us to discover two new phenomena. The first was wholesale condensation at the heater end of the device which we believe serves as the source for fluid accumulation at that end. We believe the condensation is due to disjoining pressure effects and validates one of the earliest, scientific justifications for running CVB-1; trying to understand the effects of intermolecular forces on change of phase heat transfer.

8. The high heat data has also led to a self-organizing, reverse jet that forms on the flat surfaces of the CVB (CVB-1) near the heater end. We showed this is the analog of a terrestrial rip current. The rip current is generated by the collision of Marangoni and capillary forces that, like an automated pitching machine, uses two counterrotating "vortices" that suck fluid in and thrusts it back out toward the heater. This phenomenon actually increased the surface area available for evaporative heat transfer and so the CVB generated its own geometry as the heat input increased.

9. Based on an idea that condensation at the heater end was driven by instability in the thin film coating the devices walls, we were able to show theoretically that this was the case and that the oscillation is violent enough to promote the condensation we observe. This change was shown to be a mechanism to enhance evaporative heat transfer. Periodic condensation relieved the disjoining pressure effects that kept the thin film from evaporating and allowed for more efficient heat transfer.

ANNUAL REPORTING JULY 2019

SUMMARY: During the course of this year we have continued to work on analyzing the data from both CVB-2 and CVB-1 flight experiments. We have focused most recently on experiments that involved lowering the cooler temperature on CVB-1 and trying to understand the nature of what we now feel is a rip current that is generated from the central drop as the power input is increased. We have also been rebuilding our ground-based CVB system and refining our image analysis procedures to extract more data from the images and reliably determine values for the in-situ Hamaker constant of pure fluids and fluid mixtures. We submitted a paper on the rip current. We are now trying to expand our meniscus oscillation model to partially wetting fluids and fluid mixtures. We continue to present the information at as many conferences as we could reasonably attend and have been pursuing some ideas on wicking structures and surface modification techniques using atomic layer deposition and atom-transfer-radical polymerization.

YEARLY HIGHLIGHTS

1. We spent a considerable amount of time regressing and analyzing the data from the high heat input experiments run aboard the International Space Station (ISS) in late 2010 using the 30 and 40 mm module CVB1 experiments on the ISS. We were particularly interested in the rip current that forms at high heat inputs and that disappears when we lower the cooler temperature. We have a mechanism and have submitted a paper to Applied Energy.

2. We have been analyzing data from CVB-1 that involves what happens as we lower the cooler temperature. We believe we understand what happens and why the Marangoni fingers continue to grow when the cooler temperature is reduced. Much of that work is contained in the paper to Applied Energy, but some details can be forged into another paper for this coming year.

3. We continue to work on a project from NASA to use the Physical Sciences Informatics database to continue analyzing and modeling results from CVB1. The Principal Investigator (PI) on that proposal is Vladimir Ajaev of Southern Methodist University. Based on that work we are looking at a new mechanism to describe the Marangoni fingers we observe namely, a form of hydraulic jump that occurs in thin films.

4. National Science Foundation (NSF)/CASIS project is going along but is very slow. We have rebuilt some of the experimental apparatus for ground based testing and our first task is to verify how a Hamaker constant behaves as the fluid composition changes. The pentane/isohexane mixtures we use in CVB should behave as ideal mixtures, and we will check that out first. This should lead to another paper, especially if the mixture is somehow non-ideal.

5. We presented results at 3 meetings with two more scheduled for Fall 2019; one is Pawar, A., and Plawsky, J. "An Optical Study of 3-Phase Contact Line on Silicon Surface." To be presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), November 2019.

6. We have been experimenting with several surface modification approaches designed primarily to study the influence of the Hamaker constant on evaporation and condensation. One such effort involves using atom-transfer-radical polymerization to grow block copolymer brushes on the surface. To date we have effectively made the first coatings and will be using them for studying spreading and wetting at room temperature. We have also been developing coatings using atomic layer deposition. Here, the goal is to investigate layer thickness and the effect of multilayers on spreading, wetting, evaporation, and condensation. Target materials include hafnium dioxide and titanium dioxide. The latter can be illuminated with near UV light and the excitons generated by that process used to affect the properties of the liquid layer.

7. We have been working with new ideas on designing wicking structures. The influence came from a membrane project we worked on and is still in its infancy. The goal is to develop a connection between transport, mazes and graph theory. This could provide a unique way to designing porous structures and a framework for using machine learning to optimize those designs.

Bibliography Type: Description: (Last Updated: 10/14/2025)

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